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ENERGY MANAGEMENT SYSTEMS FOR EXTREME FAST-CHARGING SYSTEMS, AND ASSOCIATED METHODS

Abstract

A method implemented by an energy management system (EMS) to optimize energy distribution in an extreme fast-charging (XFC) system includes (1) generating temporal maps for each of a plurality of energy sources, including projected capacity and energy cost over time, (2) generating a temporal map of projected load of one or more EV chargers within the XFC system, and (3) selecting one or more of the energy sources to supply energy to the XFC system over a predetermined time period based on cost, availability, and demand forecasts. This selection, for example, dynamically balances energy costs, ensures charging reliability, and/or optimizes the XFC system to support high-power charging with minimal operational expenses.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims benefit of U.S. Provisional Patent Application No. 63/559,831, filed on Feb. 29, 2024, which is incorporated herein by reference.

BACKGROUND

[0002] Charging of electric vehicles (EVs) in the United States can be broken down into three categories, or Levels. Level 1 charging is the lowest charging level as it utilizes household wiring (which is typically a 120-volt alternating current (VAC) single-phase circuit) and can account for 2.5 to 5 miles per one hour of charge. As Level 1 charging uses common household wiring, it can be considered a baseline, emergency charging method. Level 2 charging involves a 220 VAC to 240 VAC single phase circuit, commonly associated with an electric range or a clothes dryer, and accounts for a dramatic increase in charging rate over Level 1 charging, effectively increasing the range to 30-40 miles per one hour of charge. Another advantage of Level 2 charging is that requisite supporting circuits can be frequently accommodated both at homes and businesses. As such, Level 2 charging infrastructure is becoming popular at hotels, apartments, and other locations where charging is not time sensitive. Level 3 charging, commonly referred to as DC Fast Charging that is rated at 50 kW or higher, involves a higher level of electrical service, typically a three-phase circuit that is commonplace in businesses and factories, but is not available in residential applications. These three-phase circuits are commonly 208 VAC or 480 VAC and can account for 60-80 miles of range in only 20 minutes of charging. Level 3 charging can include what has recently begun to be referred to as extreme fast-charging (XFC) that requires an EV capable of accepting over 300 kW of direct current (DC) power and usually is indicative of EV battery packs being rated at 800 volts direct current (VDC) or above. With XFC, it is possible for an EV to charge from 10 percent to 80 percent capacity in under 10 minutes, or effectively close to the equivalent duration of an internal combustion engine (ICE) vehicle refueling experience. EV charging categories may vary in other countries, but differences in EV charging speed between categories are analogous to those in the United States.

SUMMARY

[0003] Disclosed herein are energy management systems (EMSs) and associated methods which will provide an autonomous and predictive means for operating an electric vehicle (EV) charging station, such as an extreme fast-charging (XFC) EV charging station. Certain embodiments of the new EMSs employ energy arbitrage to select a most economical electric energy source from two or more electric energy sources, such as an electric power grid and a photovoltaic energy source, to charge one or more battery energy storage systems (BESSs), including of one or more batteries, of an EV charging station while ensuring that sufficient energy is available to meet EV charging station needs. For example, some embodiments of new EMSs may select a combination of electric energy sources for charging the BESSs over a particular time frame, e.g., over a particular 24-hour period, to ensure that a sufficient amount of energy is stored in the BESSs to cover anticipated loads associated with charging EVs during the time frame. Additionally, these embodiments may select among the electric energy sources to ensure that the energy stored in the BESSs is obtained at a lowest feasible cost. Certain embodiments employ stochastic programming techniques, statistical inference, and estimation methods applied to historical and real-time data on traffic patterns and an established base case of power flow from the utility to determine projected electric

energy source capacity, forecast projected electric energy source costs, predict projected EV charger load, and/or optimize selection of electric energy sources based on probabilistic modeling and optimization principles.

[0004] In an embodiment, a method operable by an EMS for operating an XFC system includes (1) generating respective temporal maps of projected capacity of a plurality of electric energy sources, (2) generating respective temporal maps of projected energy cost of the plurality of electric energy sources, (3) generating a temporal map of projected load of one or more electric vehicle chargers of the extreme fast-charging system, and (4) selecting one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over a predetermined time period at least partially based on (a) the respective temporal maps of projected capacity of the plurality of electric energy sources, (b) the respective temporal maps of projected energy cost of the plurality of electric energy sources, and (c) the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system, to provide sufficient energy for meeting the projected load of the one or more electric vehicle chargers of the extreme fast-charging system while complying with at least one additional objective.

[0005] In an embodiment, a method operable by an EMS for operating an XFC system includes (1) selecting a renewable electric energy source to charge a BESS of the extreme fast-charging system during a first time period, (2) obtaining first information indicating a projected decrease in capacity of the renewable electric energy source during a third time period that is after the first time period, and (3) in response to obtaining the first information, selecting an electric power grid energy source to charge the BESS during a second time period that is after the first time period but is before the third time period, to compensate for the projected decrease in capacity of the renewable electric energy source during the third time period.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a schematic diagram of an electrical environment including an energy management system (EMS), according to an embodiment.

[0007] FIG. 2 is a schematic diagram of one possible embodiment of a controller of the FIG. 1 electrical environment.

[0008] FIG. 3 is a schematic diagram of another possible embodiment of the controller of the FIG. 1 electrical environment.

[0009] FIG. 4 is a schematic diagram of control logic of one embodiment of the EMS of FIG. 1.

[0010] FIG. 5 is a block diagram illustrating one possible implementation of the FIG. 4 EMS.

[0011] FIG. 6 is a schematic diagram of control logic of one embodiment of a load projection module of the FIG. 4 EMS.

[0012] FIG. 7 is a schematic diagram of control logic of one embodiment of a renewable projection module of the FIG. 4 EMS.

[0013] FIG. 8 is a schematic diagram of control logic of one embodiment of a grid projection module of the FIG. 4 EMS.

[0014] FIG. 9 illustrates one example of operation of the FIG. 4 EMS.

[0015] FIG. 10 is a flow chart of a method operable by an EMS for operating an extreme fast-charging (XFC) system, according to an embodiment.

[0016] FIG. 11 is a flow chart of another method operable by an EMS for operating an XFC system, according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0017] While extreme fast-charging (XFC) has significant benefits, XFC requires delivery of a large amount of energy to an electric vehicle (EV) in a short amount of time. As such, XFC

requires a high-capacity electric power source. For example, an XFC system must be capable of providing approximately 360 kilowatts of power to an EV to achieve EV charging times that are comparable to refueling times of internal combustion engine (ICE) vehicles. Such a requirement for a high-capacity electric power source is particularly acute in EV charging stations including multiple EV chargers where multiple vehicles may simultaneously undergo XFC.

[0018] EV charging stations are conventionally powered by an electric power grid, such as by an electric power grid operated by an electric utility. Electric power purchased from an electric power grid may be costly, especially during times of peak electric power demand, as electric utilities are increasingly implementing variable, e.g., time-of-day, pricing for electric energy. Additionally, while electric power grids are conventionally considered reliable electric energy sources, available electric power grid capacity may significantly vary, such as according to current load on the electric power grid, rolling brownouts, unplanned brownouts, etc. As such, an EV charging station powered solely from an electric power grid typically has high operating costs and may be incapable of supporting simultaneous XFC of multiple EVs.

[0019] A renewable electric energy source, such as a photovoltaic energy source or a wind turbine energy source, may supplement an electric power grid at an EV charging station. However, capacity of renewable energy sources may significantly vary. For example, photovoltaic energy source capacity varies by time of day, sunlight obscuration by clouds, precipitation, airborne particulates, temperature, and humidity. As such, conventional use of a renewable electric energy source to supplement an electric power grid at an EV charging station leaves the EV charging station dependent on a costly, and potentially unreliable, electric power grid for significant durations. For example, consider an EV charging station that is primarily powered from an electric power grid and includes a photovoltaic energy source for supplemental electric power. The EV charging station will be completely dependent on the electric power grid at times when the photovoltaic energy source is not producing electric power, such as at night and on days when the photovoltaic energy source is covered by snow.

[0020] Disclosed herein are energy management systems (EMSs) and associated methods which at least partially overcome the above-discussed problems by providing an autonomous and predictive means for operating an EV charging station. Certain embodiments of the new EMSs employ energy arbitrage to select a most economical electric energy source from two or more electric energy sources, such as an electric power grid and a photovoltaic energy source, to charge one or more battery energy storage systems (BESSs) of an EV charging station while ensuring that sufficient energy is available to meet EV charging station needs. For example, some embodiments of new EMSs may select a combination of electric energy sources for charging the BESSs over a particular time frame, e.g., over a particular 24-hour period, to ensure that a sufficient amount of energy is stored in the BESSs to cover anticipated loads associated with charging EVs during the time frame. Additionally, these embodiments may select among the electric energy sources to ensure that the energy stored in the BESSs is obtained at a lowest feasible cost. Energy cost may be calculated by the EMSs and/or obtained from an external source accessible to the EMSs. For example, the cost of producing energy via a renewable energy source may be calculated, and the cost of energy from an electric power grid may be obtained from an external information source.

[0021] Many variations in electric energy source capacity and/or cost are due to cyclic changes, such as seasons, and other variations can be predicted a day or two in advance, such as by weather modeling. Particular embodiments of the new EMSs generate a plurality of temporal maps, such as maps representing the capacity of each electric energy source as a function of time, the cost of each electric energy source as a function of time, and the consumption of an EV station as a function of time, for both present operation and near-term future operation. These EMS embodiments then compare the maps to determine a most cost-effective selection of electric energy sources for charging one or more BESSs of the EV station, as well as to ensure that the BESSs will be sufficiently charged to meet anticipated electric power demand of the EV station. For example, in

some embodiments, a temporal map of predicted consumption is overlaid with a temporal map of electric energy source capacity to ensure that there is sufficient electric energy source capacity to meet projected EV station load, and low-cost electric energy sources (e.g., renewable energy sources) of the available electric energy sources are selected to meet the projected load before selecting high-cost electric energy sources (e.g., an electric power grid). In certain embodiments, any projected shortfall in renewable energy source capacity over a particular time frame (e.g., over a particular 24-hour period) is compensated by (i) purchasing energy from an electric power grid at times when electric power grid rates are low and (ii) storing the purchased energy in one or more BESSs for use in powering the EV charging station over the particular time frame.

[0022] Furthermore, certain embodiments of the new EMSs are configured to determine trends in individual EV charging, as well as trends in collective EV charging, such as by (i) analyzing data of historical EV charging station operation and/or (ii) data obtained from EVs being charged, e.g., via datalinks between the EVs and EV chargers. The new EMSs may be configured to use these trends to estimate impacts of holidays, and other events, on estimated EV station consumption for a given time frame. Certain embodiments of the new EMSs use just-in-time prediction algorithms based on recent hours of operational data and historically established patterns to prioritize available energy storage within a BESS, account for the real-time status of charging stations and anticipated hours of EV station charging, and dynamically adjust the rate of solar power harvest while managing supplemental energy injection from the electrical utility. These methods facilitate the estimation of load demands of EV charging, electric energy source capacity, electric energy source costs, and/or the optimal selection of electric energy sources based on real-time and historical data-driven decision-making.

[0023] FIG. 1 is a schematic diagram of an electrical environment **100** including an EMS **102**, where EMS **102** is one embodiment on the new EMSs disclosed herein. EMS **102** is part of an XFC system **104**, which is implemented at an EV charging station **106**. It is understood, though, that EMS **102** could be used in other charging systems, including charging systems that are not capable of XFC. Additionally, EMS **102** could be separate from, but communicatively coupled to, XFC system **104** without departing from the scope hereof. Electrical environment **100** further includes a photovoltaic (PV) energy source **108** and an AC electric power grid **110**. Electrical environment **100** optionally also includes one or more other renewable energy sources **112** and an associated power disconnect switch **114**. While PV energy source **108**, other renewable energy sources **112**, and power disconnect switch **114** are depicted as being separate from XFC system **104**, in some alternate embodiments, one or more of these elements are partially or fully incorporated with XFC system **104**. For example, in some alternate embodiments, PV energy source **108** is incorporated with XFC system **104**, such as when PV energy source **108** is physically located at EV charging station **106**.

[0024] XFC system **104** includes, in addition to EMS **102**, a controller **116**, BESSs **118**, EV chargers **120**, a main DC electric power bus **122**, a respective DC/DC converter **124** for each EV charger **120**, an AC energy source **126**, a low-voltage data control line **128**, an auxiliary power system **130**, a DC/DC converter **132**, a DC electric power bus **134**, a respective DC electric power bus **136** for each BESS **118**, a DC electric power bus **138**, and a respective DC electric power bus **140** for each EV charger **120**. In this document, specific instances of an item may be referred to by use of a numeral in parentheses (e.g., BESS **118**(1)) while numerals without parentheses refer to any such item (e.g. BESSs **118**). XFC system **104** optionally further includes a DC/DC converter **142** and a DC electric power bus **144** associated with other renewable energy sources **112**. While XFC system **104** is depicted as including two BESSs **118**, the quantity of BESSs **118** of XFC system **104** may vary. Additionally, although XFC system **104** is depicted as including two EV chargers **120**, the quantity of EV chargers **120** may also vary.

[0025] XFC system **104** is designed to utilize state-of-the-art and future battery technologies capable of reaching significant charge capacity in a short amount of time, a feature that is in

significant demand within the emerging EV market. For example, as discussed below, in particular embodiments, EMS **102** is configured to control XFC system **104** to ensure that power provided to an EV is the most economical to the consumer by using renewable energy, e.g., PV energy source **108** and/or other renewable energy sources **112**, as a primary energy source and only using AC energy source **126** as a backup electric energy source, and by ensuring that system losses by unnecessary rectification/inverting functions are eliminated. Thus, in certain embodiments, the bulk of XFC system **104** consists of DC electric power buses, with AC energy source **126** providing electric energy as a backup only. However, it is understood that XFC system **104** could be modified to include additional elements without departing from the scope hereof.

[0026] PV energy source **108** includes a PV array **146**, optimizer circuitry **148**, a combiner box **150**, and a power disconnect switch **152**. In one embodiment, PV array **146** is sized to provide sufficient electric power to XFC system **104**, e.g., to meet anticipated demand of EV chargers **120**. Output of PV array **146** is optimized by optimizer circuitry **148** that includes a maximum peak power tracker (MPPT), sometimes alternately referred to as a maximum power point tracker, and associated circuitry, to help ensure maximum PV array **146** output by adjusting its input impedance such that PV array **146** operates at its maximum power point. Combiner box **150** is used to obtain a single power lead from PV energy source **108** into XFC system **104**, such as by combining electrical outputs of one or more optional additional PV arrays (not shown) with the output of PV array **146**. Combiner box **150** is electrically coupled to DC/DC converter **132** by power disconnect switch **152**. The configuration of PV energy source **108** may vary as long as it is capable of providing DC electric power to XFC system **104**. Additionally, some functionality of PV energy source **108** may be incorporated in XFC system **104**. For example, in particular alternate embodiments, DC/DC converter **132** can perform MPPT and optimizer circuitry **148** is therefore omitted.

[0027] Electric power provided by PV energy source **108** can be energized and de-energized by power disconnect switch **152**. When power disconnect switch **152** is closed, electric power generated by PV energy source **108** goes through a high-efficiency DC/DC converter **132** to match its voltage output to that necessary to power XFC system **104**. Specifically, DC/DC converter **132** converts a voltage magnitude $V_{sub.PV}$ of electric power generated by PV array **146** to a voltage magnitude $V_{sub.PV_C}$ on DC electric power bus **134**, where DC electric power bus **134** electrically couples DC/DC converter **132** to controller **116**. DC/DC converter **132** is optional if voltage magnitude $V_{sub.PV}$ is already matched to XFC system **104**. Other renewable energy sources **112** can also be incorporated to provide the necessary power to XFC system **104**. Other renewable energy sources **112** can include, but are not limited to, wind, geothermal, and hydrodynamic electric energy sources, provided that this power is predominantly DC (or is converted to DC before reaching XFC system **104**). Similar to PV energy source **108**, energizing and de-energizing other renewable energy sources **112** goes through a power disconnect switch **114** and then through DC/DC converter **142** to match the necessary voltage of XFC system **104**. Specifically, DC/DC converter **142** converts a voltage magnitude $V_{sub.O}$ of electric power generated by other renewable energy sources **112** to a voltage magnitude $V_{sub.O_C}$ on DC electric power bus **144**, where DC electric power bus **144** electrically couples DC/DC converter **142** to controller **116**. In some alternate embodiments, DC/DC converter **142** is omitted, such as if voltage magnitude V_o is already matched to the voltage of XFC system **104**.

[0028] AC energy source **126** includes a power disconnect switch **154**, an isolation transformer **156**, and a power conversion system (PCS) **158**. PCS **158** is electrically coupled to AC electric power grid **110** via the series combination of power disconnect switch **154** and isolation transformer **156**. AC electric power grid **110** provides a backup source of electric power to XFC system **104** via AC energy source **126**. Electric power from AC electric power grid **110** is likely to be more expensive than electric power from PV energy source **108** and other renewable energy sources **112**, but AC electric power grid **110** may provide capacity for charging BESSs **118**, or

directly powering XFC system **104**, when PV energy source **108** and other renewable energy sources **112** (if present) are unavailable or have insufficient capacity. After energizing/de-energizing via power disconnect switch **154**, isolation transformer **156** is employed to protect AC electric power grid **110** as well as the equipment of XFC system **104** connected to AC electric power grid **110**. Electric power from isolation transformer **156** is provided to PCS **158** that rectifies the AC electric power to DC electric power having a voltage magnitude $V_{sub.PCS}$ and is matched, for example, to the needs of BESSs **118**. Accordingly, PCS **158** is powered from AC electric power grid **110** via isolation transformer **156** and power disconnect switch **154**. PCS **158** is electrically coupled to controller **116** via DC electric power bus **138**. In some alternate embodiments, AC electric power grid **110** is replaced with a DC power grid, and AC energy source **126** is replaced with equipment, such as a DC/DC converter, to electrically couple DC electric power bus **138** with the DC power grid.

[0029] BESSs **118** serve as DC electric energy storage systems in XFC system **104** and are capable of powering XFC system **104**. Within a 24-hour cycle, BESS **118** charging options may vary depending upon conditions, as controlled by EMS **102**. Each BESS **118** is electrically coupled to controller **116** via a respective DC electric power bus **136**. In particular embodiments, each BESS **118** includes one or more high-capacity batteries, e.g., a Lithium-ion batteries, where each battery includes one or more electrochemical cells. For example, in certain embodiments, each BESS includes one or more strings of batteries electrically coupled in series. Each BESS **118** may include additional equipment, such as a charging controller, power conversion equipment, telemetry equipment, safety devices, climate control equipment, etc., in addition to batteries. Each BESS **118** provides DC electric power having a respective voltage magnitude V_B on its respective DC electric power bus **136**, where voltage magnitudes V_B are designed to meet XFC system **104** requirements. BESSs **118** could be supplemented by, or replaced with, other DC energy storage systems, including but not limited to inertial flywheel storage units.

[0030] Electric power from AC electric power grid **110** via AC energy source **126**, electric power from PV energy source **108**, and electric power from other renewable energy sources **112**, is provided to controller **116** via DC electric power buses **138**, **134**, and **144**, respectively, and the provided electric power is available for charging BESSs **118** and/or powering EV chargers **120**. Controller **116** includes, for example, one or more switching devices, such as relays, transistors, contactors, or the like (not shown), that are configured to selectively electrically couple one or more electric energy sources, e.g., PV energy source **108**, other renewable energy sources **112**, and/or AC energy source **126**, to one or more loads, e.g., BESSs **118** and/or EV chargers **120**, such as in response to signals from EMS **102**. Main DC electric power bus **122** exits controller **116** and is electrically coupled to one or more EV chargers **120**, each with an optional DC/DC converter **124** to match voltage magnitude $V_{sub.DC}$ of main DC electric power bus **122** to respective DC-only inputs **160** of EV chargers **120**. Specifically, each DC/DC converter **124** is electrically coupled to controller **116** via main DC electric power bus **122**, and each EV charger **120** is electrically coupled to its respective DC/DC converter **124** via its respective DC electric power bus **140**. Each DC/DC converter **124** is configured to convert electric power voltage magnitude $V_{sub.DC}$ on main DC electric power bus **122** to a respective electric power voltage magnitude $V_{sub.CS}$ on its respective DC electric power bus **140**.

[0031] In certain embodiments, voltage magnitude $V_{sub.PV_C}$, voltage magnitude $V_{sub.O_C}$, voltage magnitude $V_{sub.PCS}$, and voltage magnitudes $V_{sub.B}$ are each equal to voltage magnitude $V_{sub.DC}$, such that controller **116** need not perform voltage magnitude transformation, thereby promoting high efficiency and simplicity by eliminating the need for voltage magnitude transformation in controller **116**. For example, FIG. 2 is a schematic diagram of a controller **200**, which is one possible embodiment of controller **116** that does not perform voltage magnitude transformation. Controller **200** includes a switching device **202**, a switching device **204**, a switching device **206**, a switching device **208**, a switching device **210**, and an internal DC electric

power bus **212**. Each switching device **202-210** includes, for example, one or more relays, transistors, contactors, or the like. Switching device **202** is electrically coupled between DC electric power bus **134** and internal DC electric power bus **212**, and switching device **202** is controlled by a control signal $\Phi 1$. Switching device **204** is electrically coupled between DC electric power bus **136(1)** and internal DC electric power bus **212**, and switching device **204** is controlled by a control signal $\Phi 2$. Switching device **206** is electrically coupled between DC electric power bus **136(2)** and internal DC electric power bus **212**, and switching device **206** is controlled by a control signal $\Phi 3$. Switching device **208** is electrically coupled between DC electric power bus **138** and internal DC electric power bus **212**, and switching device **208** is controlled by a control signal $\Phi 4$. Switching device **210** is electrically coupled between DC electric power bus **144** and internal DC electric power bus **212**, and switching device **210** is controlled by a control signal $\Phi 5$. Internal DC electric power bus **212** is electrically coupled to main DC electric power bus **122**. EMS **102** is configured to generate control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$, and low voltage data control line **128** transmits control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$ from EMS **102** to controller **200**. In some alternate embodiments, low voltage data control line **128** is replaced with, or supplemented by, other communication equipment, such as an optical communication equipment or wireless communication equipment.

[0032] EMS **102** is configured to generate control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$ to control flow of electric power in XFC system **104**. For example, EMS **104** may cause electric power from PV energy source **108** to flow to BESSs **118** and to EV chargers **120**, for charging BESSs **118** and powering EV chargers **120**, respectively, by generating control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$ such that (i) each of switching devices **202**, **204**, and **206** is closed and (ii) each of switching devices **208** and **210** is open. As another example, EMS **102** may cause electric power from AC energy source **126** to flow to BESSs **118** and to EV chargers **120**, for charging BESSs **118** and powering EV chargers **120**, respectively, by generating control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$ such that (i) each of switching devices **204**, **206**, and **208** is closed and (ii) each of switching devices **202** and **210** is open. As an additional example, EMS **102** may cause electric power from BESSs **118** to flow to EV chargers **120**, for powering EV chargers **120** without charging BESSs **118**, by generating control signals $\Phi 1$, $\Phi 2$, $\Phi 3$, $\Phi 4$, and $\Phi 5$ such that (i) each of switching devices **204** and **206** is closed and (ii) each of switching devices **202**, **208**, and **210** is open.

[0033] Controller **200** could be modified to support additional functionality. For example, controller **200** could be modified to include one or more additional switching devices to enable simultaneous charging of BESSs **118** from one of PV energy source **108** and AC energy source **126** while powering EV chargers **120** from the other of PV energy source **108** and AC energy source **126**. As another example, controller **200** could be modified to include additional switching devices to enable simultaneous charging of BESS **118(1)** and discharge of BESS **118(2)** into EV chargers **120**. As a further example, controller **200** could be modified to include one or more devices for performing voltage magnitude conversion. For example, FIG. 3 is a schematic diagram of a controller **300**, which is one possible embodiment of controller **116** that is capable of performing voltage magnitude transformation. Controller **300** is like controller **200** except that controller **300** further includes a DC/DC converter **314** electrically coupled between internal DC electric power bus **212** and main DC electric power bus **122**. DC/DC converter **314** is configured to convert electric power voltage magnitude $V_{sub.int}$ on internal DC electric power bus **212** to electric power voltage magnitude $V_{sub.DC}$ on main DC electric power bus **122** in response to a control signal $\Phi 6$ generated by EMS **102**. In some embodiments, EMS **102** is configured to control DC/DC converter **314** such that DC/DC converter **314** only operates when needed. For example, assume a hypothetical scenario where each of voltage magnitude $V_{sub.PV_C}$, voltage magnitude $V_{sub.PCS}$, and voltage magnitudes $V_{sub.B}$ is equal to voltage magnitude $V_{sub.DC}$, but voltage magnitude $V_{sub.O_C}$ differs from voltage magnitude $V_{sub.DC}$. In this scenario, EMS **102** could be configured such that (i) DC/DC converter **314** operates as a voltage conversion device when

switching device **210** is closed and (ii) DC/DC converter **314** operates in a bypass mode when switching device **210** is open. The bypass mode is characterized, for example, by DC/DC converter **314** connecting internal DC electric power bus **212** to main DC electric power bus **122** without performing power conversion.

[0034] Referring again to FIG. **1**, auxiliary power system **130** includes a power disconnect switch **164** and an isolation transformer **166**. Auxiliary power system **130** is energized/de-energized via power disconnect switch **164**, and isolation transformer **166** protects AC electric power grid **110** as well as equipment of XFC **104** connected to AC electric power grid **110**. Auxiliary power system **130** may include additional elements, such as rectification circuitry (not shown) and a DC/DC converter (not shown). Auxiliary power system **130** is used, for example, when powering up XFC system **104** to provide an initial power source for elements of XFC system **104**, such as EMS **102** and controller **116**.

[0035] EMS **102** is configured to control operation of XFC system **104**, such as by controlling operation of controller **116** to select among PV energy source **108**, AC energy source **126**, and other renewable energy sources **112** (if present), to provide electric power to XFC system **104**, to (i) cause BESSs **118** to be charged with energy that is obtained at a lowest possible cost and (ii) ensure that sufficient energy is available to meet anticipated load of EV chargers **120**. In particular embodiments, EMS **102** prioritizes use of renewable energy sources over use of non-renewable energy sources in powering XFC system **104**. For example, in certain embodiments, EMS **102** is configured to control XFC system **104** such that (i) PV energy source **108** and other renewable energy sources **112** (when present) are prioritized for powering XFC system **104** and (ii) AC electric power grid **110** is used to supply any shortfall in renewable energy sources by sourcing energy from AC energy source **126** at times when (1) AC electric power grid **110** rates are low and (2) AC electric power grid **110** has sufficient capacity to charge BESSs **118** and/or directly power EV chargers **120**. In some embodiments, EMS **102** controls XFC system **104** at least partially based on data from external data sources **162**. FIGS. **4-9**, discussed below, illustrate examples of configuration and operation of EMS **102**. It is understood, however, that the configuration and operation of EMS **102** is not limited to the examples of FIGS. **4-9**.

[0036] FIG. **4** is schematic diagram **400** of control logic of an EMS **402**, where EMS **402** is one embodiment of EMS **102** of FIG. **1**. Schematic diagram **400** further includes PV energy source **108**, controller **116**, BESSs **118**, EV chargers **120**, AC energy source **126**, and external data sources **162**, in addition to EMS **402**. Other renewable energy sources **112** are not shown in FIG. **4** for illustrative clarity, but it is understood that other renewable energy sources **112** could be controlled by EMS **402** in a manner similar to how EMS **402** controls PV energy source **108**. Control signal flow is indicated by a light arrow, data transfer is indicated by dashed arrows, and power transfer is indicated by heavy arrows, in schematic diagram **400**. EMS **402** includes a renewable projection module **404**, a grid projection module **406**, a load projection module **408**, a system alerts module **410**, and a multi-day source selection module **412**.

[0037] External data sources **162** include, for example, control signals from one or more servers for remote control of XFC system **104**, autonomous polling or pushing of data signals from free services, and/or autonomous polling or pushing of data signals from paid services. Examples of possible data represented by signals from external data sources **162** include, but are not limited to, weather reports (reporting present weather and/or predicted weather), natural and man-made environmental conditions (present and/or predicted), traffic information, data streams from services that can monitor individual EV charging history, and/or data from AC electric power grid **110** operations, including possible sub-optimum capacity alerts or outage notifications.

[0038] Renewable projection module **404** is configured to generate (i) a respective map **414** for each renewable electric energy source (e.g., PV energy source **108** and/or other renewable energy sources **112**) representing present and projected capacity of the renewable electric energy source as a function of time, and (ii) a respective map **415** for each renewable electric energy source

representing present and projected cost of the renewable electric energy source as a function of time. In some alternate embodiments, capacity and cost of each renewable electric energy source are combined into a common temporal map for the renewable electric energy source. In certain embodiments, each map **414** is a data structure of capacity values as a function of time values, and each map **415** is a data structure of cost values as a function of time values. FIG. **4** assumes that other renewable energy sources **112** are omitted, and FIG. **4** therefore only shows two maps being generated by renewable projection module **404**, i.e., a map **414(1)** representing present and projected capacity of PV energy source **108** and a map **415(1)** representing present and projected cost of PV energy source **108**.

[0039] Grid projection module **406** is configured to generate (i) a map **416** representing present and projected capacity of AC electric power grid **110** as a function of time and (ii) a map **417** representing present and projected cost of energy purchased from AC electric power grid **110** as a function of time. In some embodiments, map **416** is a data structure of capacity values as a function of time values, and map **417** is a data structure of cost values as a function of time values. Load projection module **408** is configured to generate a map **418** of present and projected load of EV chargers **120** as a function of time. In some embodiments, map **418** is data structure of load values as a function of time values. Load projection module **408** is optionally also configured to generate a map (not shown) of present and projected cost associated with load of EV chargers **120** as a function of time.

[0040] System alerts module **410**, in turn, is configured to generate system alerts **420** for multi-day source selection module **412**. System alerts **420** are used by multi-day source selection module **412**, for example, to alter priority of electric energy source selection in response to one more conditions internal to XFC system **104** and/or conditions external to XFC system **104**. For example, system alerts **420** may reflect health of XFC system **104**, and EMS **402** may consider health of XFC system **104** when determining electric energy source priority selection.

[0041] Maps **414**, maps **415**, map **416**, map **417**, map **418**, and system alerts **420** are fed into multi-day source selection module **412**, and multi-day source selection module **412** generates one or more selection maps **422** based on these items. For example, multi-day source selection module **412** may overlay maps **414**, maps **415**, map **416**, map **417**, map **418** to find a lowest-cost selection of electric energy sources that meets the projected load of map **418**, and multi-day source selection module **412** may generate one or more selection maps **422** indicating the lowest-cost selection. As another example, multi-day source selection module **412** may provide map **414**, map **415**, map **416**, map **417**, map **418** to a correlation inference engine, which determines a lowest-cost selection of electric energy sources that meets the projected load of map **418**. The multi-day source selection module **412** may then generate one or more selection maps **422** indicating the lowest-cost selection. Selection maps **422** include, for example, a map that specifies how XFC system **104** is to be powered, as a function of time for the present and near-term future. For example, in certain embodiments, one or more selection maps **422** indicate (i) what energy source, if any, charges BESSs **118** as a function of time, and (ii) how EV chargers **120** are powered, e.g., by one or more BESSs and/or by one or more energy sources, as a function of time.

[0042] EMS **402** uses selection maps **422** to specify selected energy sources for charging BESSs **118** and/or directly powering EV chargers **120** at a lowest feasible cost, while ensuring that adequate energy is available to meet present and predicted consumption of EV chargers **120**. EMS **402** selects an electric energy source for charging BESSs **118** by controlling controller **116** to electrically couple the electric energy source to BESSs **118**. For example, in embodiments where controller **116** is embodied by controller **200** of FIG. **2**, EMS **402** selects PV energy source **108** for charging BESS **118(1)** by causing switching devices **202** and **204** to be closed, thereby electrically coupling PV energy source **108** to BESS **118(1)**. As another example, EMS **402** selects AC energy source **126** for charging BESS **118(2)** by causing switching devices **206** and **208** to be closed, thereby electrically coupling AC energy source **126** to BESS **118(2)**. As a further example, EMS

402 selects BESS **118(1)** to power EV chargers **120** by causing switching device **204** to be closed, thereby electrically coupling BESS **118(1)** to EV chargers **120**. As an additional example, EMS **402** selects PV energy source **108** for directly powering EV chargers **120** by causing switching device **202** to be closed, thereby electrically coupling PV energy source **108** to EV chargers **120**.

[0043] In some embodiments, each of renewable projection module **404**, grid projection module **406**, load projection module **408**, system alerts module **410**, and multi-day source selection module **412** is implemented by analog and/or digital electronic circuitry. For example, FIG. 5 is block diagram of an EMS **500**, where EMS **500** is one possible embodiment of EMS **402**. EMS **500** includes a processing subsystem **502**, a storage subsystem **504**, and an interface subsystem **506**. Processing subsystem **502** is communicatively coupled to each of storage subsystem **504** and interface subsystem **506**. Although processing subsystem **502**, storage subsystem **504**, and interface subsystem **506** are depicted as being different respective elements, each of these elements could be embodied by two or more sub elements that need not be collocated. For example, one or more of processing subsystem **502** and storage subsystem **504** could be at least partially embodied by a distributed computing system, such as a cloud computing system. Additionally, two or more of processing subsystem **502**, storage subsystem **504**, and interface subsystem **506** could be partially or fully combined. Interface subsystem **506** provides an interface between EMS **500** and external elements. For example, low voltage data control line **128** is communicatively coupled to EMS **500** via interface subsystem **506**, and EMS **500** is communicatively coupled to external data sources **162** via interface subsystem **506**.

[0044] Storage subsystem **504** stores information, such as instructions and/or data, for use by EMS **500**. For example, processing subsystem **502** is depicted as storing multi-day source selection module instructions **508**, renewable capacity projection instructions **510**, load capacity projection instructions **512**, grid capacity projection instructions **514**, system alert instructions **516**, maps **414**, maps **415**, map **416**, map **417**, map **418**, system alerts **420**, one or more selection maps **422**. Processing subsystem **502** is configured to execute multi-day source selection module instructions **508**, renewable capacity projection instructions **510**, load capacity projection instructions **512**, grid capacity projection instructions **514**, and system alert instructions **516** to implement multi-day source selection module **412**, renewable projection module **404**, load projection module **408**, grid projection module **406**, and system alerts module **410**, respectively.

[0045] FIG. 6 is a schematic diagram **600** of control logic of a load projection module **602**, where load projection module **602** is one embodiment of load projection module **408** of FIG. 4. Schematic diagram **600** includes system alerts module **410** and external data sources **162**, in addition to load projection module **602**. Control signal flow is indicated by a light arrow, and data transfer is indicated by dashed arrows, in schematic diagram **600**. Load projection module **602** includes a multi-day load capacity module **604**, cumulative site data **606**, vehicle specific data **608**, and date specific adjustments **610**. In some embodiments, multi-day load capacity module **604** is implemented by analog and/or digital electronic circuitry. For example, in certain embodiments, multi-day load capacity module **604** is at least partially implemented by processing subsystem **502** of FIG. 5 executing load capacity projection instructions **512** of FIG. 5. Additionally, in some embodiments, cumulative site data **606**, vehicle specific data **608**, and date specific adjustments **610** are stored in storage subsystem **504** of FIG. 5.

[0046] Multi-day load capacity module **604** is configured to generate map **418** from cumulative site data **606**, vehicle specific data **608**, date specific adjustments **610**, external data sources **162**, and system alerts **612** from system alerts module **410**. Map **418** represents, for example, several days of predicted load of EV chargers **120** from the present to the near future. Multi-day load capacity module **604** uses cumulative site data **606**, for example, to help ensure that map **418** reflects load history of EV chargers **120**. In particular embodiments of load projection module **602**, data from external data sources **162** is combined with cumulative site data **606** that represents data that is collected about the performance of a specific location (e.g., location of EV charging station **106**),

such as traffic patterns in the vicinity of the location, to ultimately provide year-to-year data to improve predictive accuracy. Cumulative site data **606** may be collected locally from EV charging station **106**. Additionally or alternately, cumulative site data **606** may be remotely collected and stored until needed.

[0047] Multi-day load capacity module **604** uses vehicle specific data **608**, for example, to further tailor projected load of EV chargers **120** to information provided by vehicles being charged by XFC system **104**. EV chargers **120** obtain vehicle specific data **608** from EVs via Wi-Fi, Bluetooth, or another communication means, during EV charging. Vehicle specific data **608** is collected, stored, and analyzed via external data sources **162** to determine trends for repeat customers that can then be utilized to provide a more personalized charging experience, while also enabling multi-day load capacity module **604** to project next visits from a given customer and a possible charging level required from said customer.

[0048] Multi-day load capacity module **604** uses date-specific adjustments **610** when determining projected load of EV chargers **120**, such as by adjusting projected load in response to holidays or data-specific changes in traffic patterns that deviate from normal. Date-specific adjustments **610** are particularly important for accounting for holidays or other events that do not maintain a set date, when predicting load of EV chargers **120**. System alerts **612** represent, for example, one or more aspects of health of XFC system **104**, such as BESS **118** health, BESS **118** capacity, or abnormal system-level energy losses. Multi-day load capacity module **604** uses data representing health of XFC system **104**, for example, to account for any health aspect of XFC system **104** that might affect predicted load of EV chargers **120**.

[0049] FIG. 7 is a schematic diagram **700** of control logic of a renewable projection module **702**, where renewable projection module **702** is one embodiment of renewable projection module **404** of FIG. 4. Schematic diagram **700** includes system alerts module **410** and external data sources **162**, in addition to renewable projection module **702**. Control signal flow is indicated by a light arrow, and data transfer is indicated by dashed arrows, in schematic diagram **700**. Renewable projection module **702** includes a multi-day renewable capacity module **704**, current weather data **706**, projected weather data **708**, and environmental adjustments **710**. In some embodiments, multi-day renewable capacity module **704** is implemented by processing subsystem **502** of FIG. 5 implementing renewable capacity projection instructions **510**. Additionally, in some embodiments, current weather data **706**, projected weather data **708**, and environmental adjustments **710** are stored in storage subsystem **504**.

[0050] Multi-day renewable capacity module **704** is configured to generate maps **414** and **415** from one or more of current weather data **706**, projected weather data **708**, environmental adjustments **710**, external data sources **162**, system alerts **712** from system alerts module **410**, and historical capacity of renewable energy sources (e.g., historical daily average capacity of PV energy source **108** and/or other renewable energy sources **112**). In particular embodiments, renewable energy sources, e.g., PV energy source **108** and other renewable energy sources **112** (when present), constitute a primary energy source for XFC system **104**, with emphasis on PV energy source **108**. It is believed the amortized cost of energy with an industrial scale photovoltaic system will be the most cost effective, depending upon location, compared to grid-based AC energy source **126** which may serve as a backup power source. Like all renewable energy sources, there is some variability in capacity of PV energy source **108** and other renewable energy sources **112** due to both natural and man-made conditions, so it is likely that renewable energy sources, even when sized correctly, may not have sufficient capacity to provide power to XFC system **104** based upon projected load from map **418**. Some of these variables are very predictable, such as sunrise/sunset times for photovoltaic power conversion and sun angle during the day with respect to solar panels, and while other variables can be predicted with a degree of certainty, such as weather modeling using current weather data **706** and projected weather data **708**. Particular embodiments of multi-day renewable capacity module **704** take these variables into account when projecting renewable energy capacity.

[0051] Additionally, some embodiments of multi-day renewable capacity module **704** consider environmental adjustments **710** when projecting renewable energy capacity. Environmental adjustments **710** include, for example, obscuration from particulate due to wild fires, dust/sand storms, etc. Some embodiments of multi-day renewable capacity module **704** also consider health of XFC system **104**, such as expressed by system alerts **712**, to adjust renewable energy capacity projections based on health of XFC system **104**. For example, multi-day renewable capacity module **704** may reduce projected capacity of PV energy source **108** in response to system alerts **712** indicating an impairment in DC converter **132**. Additional examples of possible system health expressed by system alerts **712** include, but are not limited to, (i) individual photovoltaic module health in PV array **146** (e.g., insolation, power, voltage, current, and temperature), (ii) collective health of multiple photovoltaic modules in PV array **146** (e.g., string power, string voltage, string current, and string temperature), and (iii) wind turbine parameters (e.g., speed, power, voltage, and current) in embodiments where other renewable energy sources **112** include a wind turbine.

[0052] FIG. **8** is a schematic diagram **800** of control logic of a grid projection module **802**, where grid projection module **802** is one embodiment of grid projection module **406** of FIG. **4**. Schematic diagram **800** includes system alerts module **410** and external data sources **162**, in addition to renewable projection module **802**. Control signal flow is indicated by a light arrow, and data transfer is indicated by dashed arrows, in schematic diagram **800**. Grid projection module **802** includes a multi-day grid capacity module **804**, current grid data **806**, projected grid data **808**, and system level adjustments **810**. In some embodiments, multi-day grid capacity module **804** is implemented by processing subsystem **502** of FIG. **5** executing grid capacity projection instructions **514** of FIG. **5**. Additionally, in some embodiments, current grid data **806**, projected grid data **808**, and system level adjustments **810** are stored in storage subsystem **504**.

[0053] Multi-day grid capacity module **804** is configured to generate maps **416** and **417** from one or more of current grid data **806**, projected grid data **808**, system level adjustments **810**, external data sources **162**, and system alerts **812** from system alerts module **410**. For example, in some embodiments, multi-day grid capacity module **804** uses as inputs one or more of (i) real-time health of AC electric power grid **110** from external data sources **162**, (ii) possible limitations on AC electric power grid **110** capacity from external data sources **162**, (iii) voltage fluctuations from current grid data **806**, (iv) grid power prices from current grid data **806**, and (v) projected grid reliability over several days from projected grid data **808**, to help determine a temporal landscape of AC electric power grid **110** capacity and availability. Additionally, certain embodiments of multi-day grid capacity module **804** consider system level adjustments **810** to determine if any system-level adjustments are necessary outside of normal operational parameters for AC electric power grid **110**. Examples of parameters which generate system level adjustments **810** include, but are not limited to, electromagnetic interference due to sunspot activities or other potential phenomena that might affect availability of AC electric power grid **110**. Some embodiments of multi-day grid capacity module **804** also consider health of XFC system **104** as expressed by system alerts **812** when determining AC electric power grid **110** capacity. For example, multi-day grid capacity module **804** may reduce projected capacity of AC electric power grid **110** in response to a problem with one or more of isolation transformer **156** and PCS **158**.

[0054] FIG. **9** illustrates one example of operation of EMS **402** of FIG. **4**, although it is understood that EMS **402** could operate in other manners. FIG. **9** includes a graph **902**, a graph **904**, a graph **906**, a timeline **908**, and a timeline **910**, which share a common time base. Graph **902** illustrates projected capacity of PV energy source **108** versus time, such as specified by map **414**, and graph **904** represents capacity of AC electric power grid **110** as a function of time, such as specified by map **416**. Graph **906** represents cost of energy purchased from AC electric power grid **110**, such as specified by map **417**. Timeline **908** illustrates operation of BESS **118(1)** as a function of time, such as specified by one or more selection maps **422**. Additionally, timeline **910** illustrates operation of BESS **118(2)** as a function of time, such as specified by one or more of selection maps

422.

[0055] Time $t_{sub.0}$ represents the present time. Accordingly, all times to right of time $t_{sub.0}$ are in the future, and all times to the left of time $t_{sub.0}$ are in the past. Capacity of PV energy source **108** is projected to be relatively low between times $t_{sub.4}$ and $t_{sub.5}$, as illustrated in graph **902**, such as due to predicted cloudy weather. Additionally, capacity of AC electric power grid **110** is projected to be abnormally low between times $t_{sub.7}$ and $t_{sub.8}$, as illustrated in graph **904**, such as due to a predicted brownout or upcoming scheduled maintenance. Cost of energy purchased from AC electric power grid **110** varies between a minimum value (Min.) and a maximum value (Max.), such as based on demand for energy from AC electric power grid **110**, as illustrated in graph **906**. While graph **906** depicts three levels of cost of energy purchased from AC electric power grid **110**, it is understood that cost of energy purchased from an AC electric power grid may be structured in other manners.

[0056] Between times $-t_{sub.4}$ and $-t_{sub.3}$, capacity of PV energy source **108** was zero. However, BESS **118(2)** was substantially charged, and EMS **402** therefore selected BESS **118(2)** to power EV chargers **120** by causing BESS **118(2)** to operate in a discharge mode **912** between times $-t_{sub.4}$ and $-t_{sub.3}$, where discharge mode **912** is characterized by BESS **118(2)** powering EV chargers **120**. BESS **118(1)** was also substantially charged at time $-t_{sub.4}$, but EMS **402** caused BESS **118(1)** to be idle between times $-t_{sub.4}$ and $-t_{sub.3}$ to save its capacity for later use. PV energy source **108** had significant capacity between times $-t_{sub.3}$ and $-t_{sub.2}$, as illustrated in graph **902**. Accordingly, EMS **402** selected PV energy source **108**, instead of AC electric power grid **110**, for charging BESS **118(2)** between times $-t_{sub.3}$ and $-t_{sub.2}$ because cost of energy from PV energy source **108** is less than cost of energy from AC electric power grid **110**. Accordingly, BESS **118(2)** operated in a PV charge mode **914** between times $-t_{sub.3}$ and $-t_{sub.2}$, which is characterized by BESS **118(2)** being charged by PV energy source **108**. Additionally, EMS **402** selected BESS **118(1)** to power EV chargers **120** between times $-t_{sub.3}$ and $-t_{sub.1}$ by causing BESS **118(1)** to operate in a discharge mode **916** during this time period, where discharge mode **916** is characterized by BESS **118(1)** powering EV chargers **120**. At time $-t_{sub.1}$, EMS **402** selected BESS **118(2)** to power EV chargers **120** by causing BESS **118(2)** to operate in a discharge mode **918**, where discharge mode **918** is characterized by BESS **118(2)** powering EV chargers **120**. Additionally, at time $-t_{sub.1}$, EMS **402** selected PV energy source **108** to charge BESS **118(1)** by causing BESS **118(1)** to operate in a PV charge mode **920** which is characterized by PV energy source **108** charging BESS **118(1)**.

[0057] At time $-t_{sub.1}$, (i) EMS **402** obtains data representing projected weather over a future time duration (e.g., several days) from external data sources **162**, (ii) estimates capacity of PV energy source **108** using renewable projection module **404**, (iii) determines projected load (not shown in FIG. **9**) of EV chargers **120** using load projection module **408**, and (iv) generates one or more selection maps **422** using multi-day source selection module **412**. In this example, the one or more selection maps **422** specify selected electric energy sources for charging BESSs **118** and powering EV chargers **120** between times $t_{sub.0}$ and $t_{sub.8}$, as illustrated in timelines **908** and **910**. In response to projected decrease in PV energy source **108** capacity between times $t_{sub.4}$ and $t_{sub.5}$ and projected drop in available capacity of AC electric power grid **110** between times $t_{sub.7}$ and $t_{sub.8}$, EMS **402** selects AC electric power grid **110** to charge BESS **118(1)** by causing BESS **118(1)** to operate in a grid charge mode **924** for between times $t_{sub.0}$ and $t_{sub.1}$, to compensate for the aforementioned decreases in capacity of PV energy source **108** and AC electric power grid **110**. Grid charge mode **924** is characterized by AC electric power grid **110** charging BESS **118(1)**. It should be noted that EMS **402** causes BESS **118(1)** to operate in grid charge mode **924** solely when cost of energy purchased from AC electric power grid **110** is low. Additionally, EMS **402** selects PV energy source **108** for charging BESS **118(1)** by causing BESS **118(1)** to operate in a PV charge mode **920** between times $-t_{sub.1}$ and $t_{sub.2}$, which is characterized by PV energy source **108** charging BESS **118(1)**. Furthermore, EMS **402** selects AC electric power grid **110** for charging

BESS **118(1)** between times t.sub.3 and t.sub.4 by causing BESS **118(1)** to operate in a grid charge mode **926**, due to (i) the upcoming decrease in capacity of AC electric power grid **110** and (ii) the relatively low cost of energy purchased from AC electric power grid **110** during this time frame. Grid charge mode **926** is characterized by BESS **118(1)** being charged from AC electric power grid **110**.

[0058] EMS **402** selects BESS **118(1)** to power EV chargers **120** between time t.sub.4 and t.sub.7 by causing BESS **118(1)** to operate in a discharge mode **928**, which is characterized by BESS **118(1)** powering EV chargers **120**. Additionally, EMS **402** prepares BESS **118(2)** for future use in powering EV chargers **120** by (i) selecting AC electric power grid **110** to charge BESS **118(2)** between times t.sub.4 and t.sub.6 by causing BESS **118(2)** to operate in a grid charge mode **930** and (ii) selecting PV energy source **108** to charge BESS **118(2)** between times t.sub.4 and t.sub.5 by causing BESS **118(2)** to operate in a PV charge mode **932** when PV energy source **108** is projected to have capacity. Grid charge mode **930** is characterized by BESS **118(2)** being charged from AC electric power grid **110**, and PV charge mode **932** is characterized by BESS **118(2)** being charged by PV energy source **108**. Charging BESS **118(2)** between times t.sub.4 and t.sub.6 from AC electric power grid **110** will require purchasing energy from AC electric power grid **110** at a relatively high price, but this purchase is required due to the projected low capacity of PV energy source **108** during this time frame and the projected upcoming drop in capacity of AC electric power grid **110**. Finally, EMS **402** selects BESS **118(2)** to power EV chargers **120** between times t.sub.7 and t.sub.8 by causing BESS **118(2)** to operate in a discharge mode **934**, which is characterized by BESS **118(2)** powering EV chargers **120**.

Control Logic

[0059] Certain embodiments of EMS **102** and EMS **402** operate according to control logic of TABLES 1-4 below. It is understood, though, that EMS **102** and EMS **402** are not limited to operating according to this control logic. The control logic of TABLES 1-4 assumes that XFC system **104** includes N charging stations, where N is a positive integer. For example, in the example of FIG. 1 where XFC system **104** includes two EV chargers **120**, N is equal to two.

TABLE-US-00001

TABLE 1	Definitions
BESS	BESSs 118
SOC	State of Charge of BESSs 118
C1	Power Demand from EV Charger 120(1)
C2	Power Demand from EV Charger 120(2)
.circle-solid.	.circle-solid. .circle-solid. .circle-solid. .circle-solid. .circle-solid.
CN	Power Demand from EV Charger 120(N)
TP	Summation of C1, C2, . . . CN
PS	Aggregate Source of Power to XFC System 104
PV	Source of Power from PV Energy Source 108
BS	Source of Power from BESSs 118
CG	Source of Power from AC Electric Power Grid 110

TABLE-US-00002

TABLE 2	EMS Priorities
1	To monitor, manage, and ensure uninterrupted dispensing of TP at any one point, i.e., PS must always be greater than or equal to TP.
2	To deliver PS based on the following priority: (1) PV, (2) BESS, and (3) CG.
3	To monitor the PV system and the SOC of the BESS.
4	To monitor and record the total electricity dispensed between PC & CG during the three monitoring periods discussed below in TABLE 4.

TABLE-US-00003

TABLE 3	PS Control Logic
1	If the sum of C1:CN is < PV, then PS = PV.
2	If the sum of C1:CN is > PV, then PS = PV + BS, where BS = PS - max of PV.
3	If TP = 0 or is < PS, then all excess PV shall recharge BESS.
4	If PV = 0, then TP = PS = BS.
5	If PV = 0 and TP > BS, then PS = CG. (Note: this scenario should not occur under normal operating circumstances and is a last resort of power source.)

TABLE-US-00004

TABLE 4	BESS Recharging Logic
1	From 07:00H to 17:00H, if SOC < 40%, the BESS shall be recharged by CG until 80%.
2	From 17:00H to 21:00H, CG shall only be on standby mode unless BESS < 10%.
3	From 21:00H to 07:00H, CG shall continuously recharge BESS until 100%.

Additional Embodiments

[0060] FIGS. 10 and 11, discussed below, illustrate several additional examples of operation of EMS **102** and EMS **402**. It is understood, though, that EMS **102** and EMS **402** are not limited to

operating according to the examples of FIGS. 10 and 11.

[0061] FIG. 10 is a flow chart of a method 1000 operable by an EMS for operating an XFC system. In a block 1002 of method 1000, respective temporal maps of projected capacity of a plurality of electric energy sources are generated. In one example of block 1002, renewable projection module 404 generates map 414(1), and grid projection module 406 generates map 416. In a block 1004 of method 1000, respective temporal maps of projected energy cost of the plurality of electric energy sources are generated. In one example of block 1004, renewable projection module 404 generates map 415(1), and grid projection module 406 generates map 417. In a block 1006 of method 1000, a temporal map of projected load of one or more EV chargers of the XFC system is generated. In one example of block 1006, load projection module 408 generates map 418.

[0062] In a block 1008 of method 1000, one or more of the plurality of electric energy sources are selected to provide energy to the XFC system over a predetermined time period at least partially based on (a) the respective temporal maps of projected capacity of the plurality of electric energy sources, (b) the respective temporal maps of projected energy cost of the plurality of electric energy sources, and (c) the temporal map of projected load of the one or more EV chargers of the extreme fast-charging system, to provide sufficient energy for meeting the projected load of the one or more EV chargers of the XFC system while complying with at least one additional objective. In one example of block 1008, multi-day source selection module 412 generates one or more maps 422 specifying the selection of electric energy sources illustrated in timelines 908 and 910 of FIG. 9, to achieve lowest cost of powering EV chargers 120 and ensuring sufficient energy to meet projected load of EV chargers 120.

[0063] FIG. 11 is a flow chart of a method 1100 operable by an EMS for operating an XFC system. In a block 1102 of method 1100, a renewable electric energy source is selected to charge a BESS of the XFC system during a first time period. In one example of block 1102, EMS 402 selects PV energy source 108 to charge BESS 118(1) between times $-t_{sub.1}$ and $t_{sub.2}$, as illustrated in FIG. 9. In a block 1104 of method 1100, the EMS obtains first information indicating a projected decrease in capacity of the renewable electric energy source during a third time period that is after the first time period. In one example of block 1104, EMS 402 obtains from external data sources 162 information indicating the projected decrease in capacity of AC electric power grid 110 between times $t_{sub.7}$ and $t_{sub.8}$, as illustrated in FIG. 9. In a block 1106 of method 1100, in response to obtaining the first information, the EMS selects an electric power grid energy source to charge the

[0064] BESS during a second time period that is after the first time period but is before the third time period, to compensate for the projected decrease in capacity of the renewable electric energy source during the third time period. In one example of block 1106, EMS 402 selects AC electric power grid 110 to charge BESS 118(1) between times $t_{sub.3}$ and $t_{sub.4}$, in response to receiving the information indicating the projected decrease in capacity of AC electric power grid 110 between times $t_{sub.7}$ and $t_{sub.8}$.

Combinations of Features

[0065] Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations. [0066] (A1) A method operable by an EMS for operating XFC system includes (1) generating respective temporal maps of projected capacity of a plurality of electric energy sources, (2) generating respective temporal maps of projected energy cost of the plurality of electric energy sources, (3) generating a temporal map of projected load of one or more electric vehicle chargers of the extreme fast-charging system, and (4) selecting one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over a predetermined time period at least partially based on (a) the respective temporal maps of projected capacity of the plurality of electric energy sources, (b) the respective temporal maps of projected energy cost of the plurality of electric energy sources, and (c) the temporal map of projected load of the one or more electric vehicle chargers of

the extreme fast-charging system, to provide sufficient energy for meeting the projected load of the one or more electric vehicle chargers of the extreme fast-charging system while complying with at least one additional objective. [0067] (A2) In the method denoted as (A1), the at least one additional objective may include minimizing cost of energy for meeting the projected load of the one or more electric vehicle chargers of the extreme fast-charging system. [0068] (A3) In either one of the methods denoted as (A1) and (A2), the at least one additional objective may include prioritizing a first electric energy source of the plurality of electric energy sources over a second electric energy source of the plurality of electric energy sources. [0069] (A4) In the method denoted as (A3), (1) the first electric energy source may be a renewable electric energy source, and (2) the second electric energy source may be a non-renewable electric energy source. [0070] (A5) In any one of the methods denoted as (A1) through (A4), generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system may include obtaining data representing historical use of energy by the extreme fast-charging system from a data source accessible to the EMS. [0071] (A6) In any one of the methods denoted as (A1) through (A5), generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system may include obtaining data associated with customers of the extreme fast-charging system from a data source accessible to the EMS. [0072] (A7) In any one of the methods denoted as (A1) through (A6), generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system may include obtaining data representing holidays at a location of the extreme fast-charging system from a data source accessible to the EMS. [0073] (A8) In any one of the methods denoted as (A1) through (A7), generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system may include obtaining data representing traffic patterns in a vicinity of the extreme fast-charging system from a data source accessible to the EMS. [0074] (A9) Any one of the methods denoted as (A1) through (A8) may further include controlling charging of one or more BESSs of the extreme fast-charging system from energy provided to the extreme fast-charging system. [0075] (A10) Any one of the methods denoted as (A1) through (A9) may further include simultaneously (a) charging a first BESS of the extreme fast-charging system from energy provided to the extreme fast-charging system and (b) discharging a second BESS of the extreme fast-charging system at least partially for powering the one or more electric vehicle chargers of the extreme fast-charging system. [0076] (A11) In any one of the methods denoted as (A1) through (A10), the EMS may be configured to select one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system further based on health of the extreme fast-charging system. [0077] (A12) In any one of the methods denoted as (A1) through (A11), the plurality of electric energy sources may include (a) a renewable electric energy source and (b) an electric power grid energy source. [0078] (A13) In any one of the methods denoted as (A1) through (A12), (1) the plurality of electric energy sources may include a photovoltaic energy source and (2) generating respective temporal maps of projected capacity of the plurality of electric energy sources may include generating a temporal map including projected capacity of the photovoltaic energy source at least partially based on one or more actual or predicted environmental conditions at a location of the photovoltaic energy source. [0079] (A14) In any one of the methods denoted as (A1) through (A13), (1) the plurality of electric energy sources may include an electric power grid energy source and (2) generating respective temporal maps of projected capacity of the plurality of electric energy sources may include generating a temporal map including projected capacity of the electric power grid energy source at least partially based on a projected reliability of the electric power grid energy source. [0080] (A15) In any one of the methods denoted as (A1) through (A14), (1) the plurality of electric energy sources may include a renewable electric energy source and an electric power grid energy source and (2) selecting the one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over the predetermined time period may include generating a first control signal causing a first switching device to be closed

between a first time and a second time, the first switching device electrically coupling the renewable electric energy source to a BESS of the extreme fast-charging system. [0081] (A16) In the method denoted as (A15), selecting the one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over the predetermined time period may further include generating a second control signal causing a second switching device to be closed between the second time and a third time, the second switching device electrically coupling the electric power grid energy source to the BESS of the extreme fast-charging system. [0082] (B1) A method operable by an EMS for operating an XFC system includes (1) selecting a renewable electric energy source to charge a BESS of the extreme fast-charging system during a first time period, (2) obtaining first information indicating a projected decrease in capacity of the renewable electric energy source during a third time period that is after the first time period, and (3) in response to obtaining the first information, selecting an electric power grid energy source to charge the BESS during a second time period that is after the first time period but is before the third time period, to compensate for the projected decrease in capacity of the renewable electric energy source during the third time period. [0083] (B2) The method denoted as (B1) may further include selecting the second time period to correspond to a time period when cost of energy purchased from the electric power grid energy source is at a minimum value. [0084] (B3) Either one of the methods denoted as (B1) and (B2) may further include powering one or more electric vehicle (EV) chargers from the BESS. [0085] (B4) In any one of the methods denoted as (B1) through (B3), (1) selecting the renewable electric energy source to charge the BESS during the first time period may include generating a first control signal causing a first switching device to be closed during the first time period, the first switching device electrically coupling the renewable electric energy source to the BESS, and (2) selecting the electric power grid energy source to charge the BESS during the second time period may include generating a second control signal causing a second switching device to be closed during the second time period, the second switching device electrically coupling the electric power grid energy source to the BESS. [0086] Changes may be made in the above methods, devices, and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

Claims

1. A method operable by an energy management system (EMS) for operating an extreme fast-charging system, the method comprising: generating respective temporal maps of projected capacity of a plurality of electric energy sources; generating respective temporal maps of projected energy cost of the plurality of electric energy sources; generating a temporal map of projected load of one or more electric vehicle chargers of the extreme fast-charging system; and selecting one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over a predetermined time period at least partially based on (a) the respective temporal maps of projected capacity of the plurality of electric energy sources, (b) the respective temporal maps of projected energy cost of the plurality of electric energy sources, and (c) the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system, to provide sufficient energy for meeting the projected load of the one or more electric vehicle chargers of the extreme fast-charging system while complying with at least one additional objective.
2. The method of claim 1, wherein the at least one additional objective comprises minimizing cost of energy for meeting the projected load of the one or more electric vehicle chargers of the extreme

fast-charging system.

3. The method of claim 1, wherein the at least one additional objective comprises prioritizing a first electric energy source of the plurality of electric energy sources over a second electric energy source of the plurality of electric energy sources.

4. The method of claim 3, wherein: the first electric energy source is a renewable electric energy source; and the second electric energy source is a non-renewable electric energy source.

5. The method of claim 1, wherein generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system comprises obtaining data representing historical use of energy by the extreme fast-charging system from a data source accessible to the EMS.

6. The method of claim 1, wherein generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system comprises obtaining data associated with customers of the extreme fast-charging system from a data source accessible to the EMS.

7. The method of claim 1, wherein generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system comprises obtaining data representing holidays at a location of the extreme fast-charging system from a data source accessible to the EMS.

8. The method of claim 1, wherein generating the temporal map of projected load of the one or more electric vehicle chargers of the extreme fast-charging system comprises obtaining data representing traffic patterns in a vicinity of the extreme fast-charging system from a data source accessible to the EMS.

9. The method of claim 1, further comprising controlling charging of one or more battery energy storage systems (BESSs) of the extreme fast-charging system from energy provided to the extreme fast-charging system.

10. The method of claim 1, further comprising simultaneously (a) charging a first battery energy storage system (BESS) of the extreme fast-charging system from energy provided to the extreme fast-charging system and (b) discharging a second BESS of the extreme fast-charging system at least partially for powering the one or more electric vehicle chargers of the extreme fast-charging system.

11. The method of claim 1, wherein the EMS is configured to select one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system further based on health of the extreme fast-charging system.

12. The method of claim 1, wherein the plurality of electric energy sources comprises (a) a renewable electric energy source and (b) an electric power grid energy source.

13. The method of claim 1, wherein: the plurality of electric energy sources comprises a photovoltaic energy source; and generating respective temporal maps of projected capacity of the plurality of electric energy sources comprises generating a temporal map including projected capacity of the photovoltaic energy source at least partially based on one or more actual or predicted environmental conditions at a location of the photovoltaic energy source.

14. The method of claim 1, wherein: the plurality of electric energy sources comprises an electric power grid energy source; and generating respective temporal maps of projected capacity of the plurality of electric energy sources comprises generating a temporal map including projected capacity of the electric power grid energy source at least partially based on a projected reliability of the electric power grid energy source.

15. The method of claim 1, wherein: the plurality of electric energy sources comprise a renewable electric energy source and an electric power grid energy source; and selecting the one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over the predetermined time period comprises generating a first control signal causing a first switching device to be closed between a first time and a second time, the first switching device electrically

coupling the renewable electric energy source to a battery energy storage system (BESS) of the extreme fast-charging system.

16. The method of claim 15, wherein selecting the one or more of the plurality of electric energy sources to provide energy to the extreme fast-charging system over the predetermined time period further comprises generating a second control signal causing a second switching device to be closed between the second time and a third time, the second switching device electrically coupling the electric power grid energy source to the BESS of the extreme fast-charging system.

17. A method operable by an energy management system (EMS) for operating an extreme fast-charging system, the method comprising: selecting a renewable electric energy source to charge a battery energy storage system (BESS) of the extreme fast-charging system during a first time period; obtaining first information indicating a projected decrease in capacity of the renewable electric energy source during a third time period that is after the first time period; and in response to obtaining the first information, selecting an electric power grid energy source to charge the BESS during a second time period that is after the first time period but is before the third time period, to compensate for the projected decrease in capacity of the renewable electric energy source during the third time period.

18. The method of claim 17, further comprising selecting the second time period to correspond to a time period when cost of energy purchased from the electric power grid energy source is at a minimum value.

19. The method of claim 17, further comprising powering one or more electric vehicle (EV) chargers from the BESS.

20. The method of claim 17, wherein: selecting the renewable electric energy source to charge the BESS during the first time period comprises generating a first control signal causing a first switching device to be closed during the first time period, the first switching device electrically coupling the renewable electric energy source to the BESS; and selecting the electric power grid energy source to charge the BESS during the second time period comprises generating a second control signal causing a second switching device to be closed during the second time period, the second switching device electrically coupling the electric power grid energy source to the BESS.
